

# Toy Model 5G Public Baseline

## Full-SPARC Close-Out and 5G-N Roadmap

V4.2 / 6D Projection Framework - 2026.05.28

|             |                                                                                             |
|-------------|---------------------------------------------------------------------------------------------|
| Status      | Full-SPARC diagnostic complete; not physical validation                                     |
| Dataset     | 175 SPARC galaxies; 3391 radial points                                                      |
| Main result | Relational metric reconstruction improves over Newtonian baryonic baseline on most galaxies |
| Next object | 5G-N - Relational Metric Generation Law                                                     |

### 1. What this package is

Toy Model 5G is the first real-data diagnostic track after the V4.2 GitHub publication baseline. It is best read as a reproducible falsification and metric-reconstruction protocol inside a speculative integrated geometric research architecture.

The public frame is: Relationship -> Organization -> Observable Order. For 5G, the observable order is the galaxy rotation curve. The tested hidden object is a relational metric distortion field, not a dark halo.

### 2. What changed after the A-to-K failure trail

Toy Models 5G-A through 5G-K tested scalar thresholds, compactness penalties, galaxy-wide mass scaling, simple nonlocal kernels, typed scheduling links, and Laplacian relaxation. These attempts produced useful diagnostics, but did not provide a stable physical engine.

The important correction was to stop modelling the residual as a direct positive acceleration and instead reconstruct the effective metric distortion:

$$\chi(R) = \ln[g_{\text{obs}}(R) / g_{\text{bar}}(R)]$$

This target can handle both observed acceleration exceeding the baryonic baseline and baryonic overshoot regions where  $g_{\text{bar}}$  is greater than  $g_{\text{obs}}$ .

### 3. Full-SPARC result

| Mode             | R2 chi | RMS 5G-M   | RMS Newtonian | Improved galaxies |
|------------------|--------|------------|---------------|-------------------|
| frozen_four      | 0.4613 | 36.46 km/s | 58.57 km/s    | 139 / 175         |
| train_groupkfold | 0.5669 | 24.69 km/s | 58.57 km/s    | 155 / 175         |

The frozen\_four mode is the harsh extrapolation test: coefficients learned from the original four-galaxy 5G-L baseline were applied to all 175 galaxies. The train\_groupkfold mode tests the same feature set with galaxy-group holdout.

### 4. Interpretation

The result is strong enough to justify the next mathematical phase. It is not strong enough to claim physical validation. The correct statement is that relational metric reconstruction outperforms the Newtonian baryonic baseline over most of the SPARC archive in this test, while systematic failures remain.

### 5. What this does not claim

- It does not prove physical 3T.
- It does not solve dark matter.
- It does not replace Lambda-CDM.
- It does not disprove particle dark matter.
- It does not address CMB peaks, Bullet Cluster, lensing, BAO, supernovae, or structure formation.

- It does not derive General Relativity, QFT, the Standard Model, or physical constants.

## 6. Why this is still important

The full-SPARC run provides a reproducible empirical reason to treat the rotation-curve residual as a metric / connection reconstruction problem rather than as another direct acceleration patch. It shifts the next work from curve-fitting to writing a relational metric generation law.

## 7. 5G-N Roadmap

5G-N should attempt to generate  $\chi(R)$  from baryonic organization, topology, and boundary conditions by one frozen relational metric law. It must use held-out galaxies, forbid per-galaxy tuning, and report failures transparently.

## 8. Publication readiness statement

This package is suitable for GitHub publication as a disciplined diagnostic baseline if the README, strict non-claims, data provenance, QA checklist, and 5G-N roadmap remain attached. It should not be published as a claim of solved dark matter.